

src/components/KnowledgeGraphPanel.tsx

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Overview

A new `KnowledgeGraphPanel` component (`src/components/KnowledgeGraphPanel.tsx`, lines 117-692) adds a UI for building, viewing, and interacting with a code knowledge graph. It introduces LLM-powered code explanations and a fullscreen explorer.

Key changes

- **Imports** – React hooks, `createPortal` (R13), type imports (`FileChange` , `LlmSettings` , `DocWorkspaceMetrics` – R14-16), and `isNovadiffDocsReservedPath` (R17).
- **Props interface** (`KnowledgeGraphPanelProps` , R26) defines comparison flags, titles, metrics, and loading states.
- **Helper functions** – `parseProgressMessage` (R62-68) extracts progress numbers; `buildAutoKey` (R70-76) creates a cache key.
- **Error boundary** (`GraphExplorerErrorBoundary` , R79-115) catches explorer render errors and offers a retry button.
- **Component logic** – state for build side, progress, stats, graph payload, viewer visibility, fullscreen, and error.
 - `useEffect` hooks listen to `window.electronAPI.knowledgeGraphProgress` , load/build graphs, defer work with `requestIdleCallback` (fallback `setTimeout`), handle fullscreen key events, and mount the explorer.
 - `runBuild` triggers a graph build, uses caching via `loadGraphFromDisk` , and updates UI state.
 - `explainCode` streams LLM explanations, saves artifacts, and integrates with the explorer.
- **UI rendering** – conditional explorer, progress card, stats grid, toolbar actions, and fullscreen portal via `createPortal` .

Impact

- Adds a fully-featured knowledge-graph panel, expanding documentation capabilities.
- Uses `requestIdleCallback` (or `setTimeout`) to defer heavy graph scans, reducing initial UI lag.
- Progress and activity are reported through `useBackgroundActivityActionsOptional` , improving user feedback.
- New types and helper functions centralize logic; the error boundary isolates explorer failures.

Risks & follow-ups

- **Electron API** – the component relies on `window.electronAPI` (e.g., `buildKnowledgeGraph` , `readKnowledgeGraph` , `llmSummarizeStream`). Verify these APIs are exposed in all builds.

- **Idle callback support** – fallback timing should be tested on environments lacking `requestIdleCallback`.
- **Portal usage** – `document.body` must exist when rendering (e.g., during SSR or tests).
- **Error boundary** – errors are logged to console; ensure the retry path resets state correctly.
- **Type safety** – run TypeScript linting to confirm imported types match actual shapes.